A scaling law of contextual persistence in human language

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Abstract

Human language exhibits lawful structure at the level of words (frequency, vocabulary growth) and word pairs (co-occurrence across distance). Here we show that the arrangement of words in sequence — a central determinant of meaning — obeys a comparable law. Using large language models as probabilistic probes, we measured the reduction in target perplexity conferred by prior context at distance d beyond that of the same words scrambled; this difference, the contextual persistence function $P(d)$, isolates the influence of arrangement. Across ten corpora spanning six language families and written and spoken modalities, $P(d)$ decayed approximately as $1/d$ ($P(d) \propto d^{-\alpha}$, mean $\alpha = 1.04$; median $r^2 = 0.96$). The effect vanished in scrambled and synthetic controls, replicated across independent probes, and its scale-free form did not appear in genomic or protein sequences under domain-native models. An exponent near 1 distributes contextual influence approximately uniformly across logarithmic timescales. The results establish a scaling law of contextual persistence in human language.

1 Introduction

Human language has a rich statistical structure, aspects of which have been captured by quantitative laws [Altmann and Gerlach, 2016, Tanaka-Ishii, 2021]. Word frequencies follow Zipf's law [Zipf, 1949], and vocabulary growth follows Heaps' law [Heaps, 1978]; both are scale-free and recur with striking consistency across languages and corpora. What makes such a regularity a law is the reproducibility of its exponent, not the power-law form alone: across a wide range of languages and corpora, Zipf's exponent sits near unity and Heaps' recurs within a characteristic sublinear band. Both of these laws describe the units in aggregate, and so occupy the most basic level of a hierarchy of statistical structure. A second level concerns the units not singly but in pairs: how their occurrences co-vary across distance. Here too a high degree of statistical structure has been uncovered: the mutual information between symbols falls off as an approximate power law of their separation, decaying far more slowly than expected under conventional finite-order Markov models [Ebeling and Pöschel, 1994, Lin and Tegmark, 2017, Montemurro and Pury, 2002].

Neither of these measures, however, isolates the order in which units appear. Unit frequencies are order-blind by construction, and long-range mutual information can reflect repeated lexical and topical content even when the local ordering of that content is disrupted. Yet the same words at the same separations can compose entirely different meanings depending on their arrangement: grammatical relations and compositional meaning live in arrangement itself, not in the inventory of words or their spacing. Representational accounts of contextual dependence, such as discourse and situation models [van Dijk and Kintsch, 1983] and resonance-based accounts of memory access during reading [Myers and O'Brien, 1998], describe it qualitatively; no lawful quantitative structure has been identified. Like co-occurrence, order-specific dependence can be posed as a question of distance: how does the probability of a given word depend on the arrangement of the other words in the sequence, and how does that dependence change with distance?

Reaching this third level requires estimating the probability of specific words given ordered combinations of other words. This is the arrangement-level analogue of the pairwise calculation underlying long-range mutual information: where mutual information asks how the occurrence of one word co-varies with that of another at a given separation, this quantity asks how the probability of a word depends on the ordered configuration of others — whether immediately adjacent, as in what follows a given construction, or far back in the sequence. Classical corpus statistics approximate it only crudely. Autoregressive transformer models [Vaswani et al., 2017, Radford et al., 2019] provide a tool for capturing this structure directly: trained only to predict the next token, they assign token-level conditional probabilities over context lengths far exceeding those tractable with conventional corpus estimators, and are sensitive to precisely the order-dependent structure that raw co-occurrence statistics cannot isolate. Used as measuring instruments, or probes, rather than as objects of study, they provide a new vantage point on this level of linguistic structure.

Here we measure this dependence directly. Like mutual information, the quantity is a statistical dependence, but it is taken between an individual word and the ordered configuration of the words that precede it rather than between pairs of words. A probe's next-token probability supplies the measure of influence: how much the probability assigned to a fixed upcoming span changes when prior context is revealed. Distance along the sequence provides the axis of measurement: revealing the true context step by step, farther and farther back, makes that influence a function of distance d.

The method of measuring next-token prediction as a function of context length has a distinguished pedigree. Shannon's guessing experiments [Shannon, 1951] asked how much uncertainty about what comes next remains as a reader is given longer and longer stretches of the preceding text, and Hilberg's conjecture [Hilberg, 1990, Debowski, 2015], a reinterpretation of Shannon's data, holds that the answer follows a power law: prediction keeps improving with context length, with no floor reached within the measured range. Our intact-context condition extends this measurement, with a probe reaching far longer contexts. Critically, unlike these earlier studies, our measure specifically addresses the role of order within the sequence. In that tradition context was always presented intact and only its length was varied, so the resulting curve is a total, mixing what is gained from the words revealed with what is gained from their arrangement, with no way to tell the two apart. At each step we therefore ask how much the revealed context reduces the probe's surprise, and subtract the reduction obtained when the same words are supplied in scrambled order — the entire revealed span permuted, so that the baseline retains vocabulary and topical content but no arrangement. What remains is the additional benefit of intact sequential arrangement: the order-specific contextual influence. Its per-token gain as context is extended past distance d defines the *contextual persistence function* (CPF), $P(d)$; its functional form we call the *contextual persistence law*. Note: we use *persistence* in a statistical sense — the continued predictive influence of ordered prior context as a function of distance — not the persistence of any internal representation or cognitive process: the function is the operational measure, and the law is the regularity it reveals. We apply it across ten corpora spanning six language families, multiple genres, and written and spoken modalities. Throughout, we treat the CPF as a law governing natural-language sequences, a property of the linguistic product; whether and how it maps onto the real-time mechanisms of human production and comprehension is a separate, empirically open question we return to in the Discussion.

We find that $P(d)$ decays approximately as $1/d$ ($P(d) \propto d^{-\alpha}$), with mean decay exponent

$\alpha = 1.04$ (SD = 0.15) and no characteristic cutoff across nearly two orders of magnitude in distance. This decay is consistent across language families, genres, and modalities, vanishes in random-token controls, scales with the integrity of sequential structure, and is broadly distributed across prior sentences rather than concentrated in a small number of anchors. The CPF describes a continuous, scale-free temporal structure in human language, measurable directly from the linguistic sequence and replicable across independently trained probe models (Llama-3.1-8B, Mistral-7B).

The shape of $P(d)$ places language alongside other human behaviours and neural processes that exhibit scale-free temporal structure [Gilden, 2001, Barabási, 2005, Linkenkaer-Hansen et al., 2001, Beggs and Plenz, 2003, Kello et al., 2010], and constrains the class of mechanisms capable of producing such structure. In particular, the absence of any characteristic scale across the measured range, combined with the graded contributions of prior content distributed across the sequence, is difficult to reconcile with architectural accounts of memory that posit a sharp division between maintained and retrieved material with distinguishable temporal dynamics. The pattern is consistent instead with mechanisms in which prior content continues to shape ongoing generation through the evolving probability structure of the sequence itself.

2 Results

2.1 The contextual persistence function

We define the contextual persistence function (CPF), $P(d)$, as the *order-specific* predictive influence of prior context at distance d from the current point of generation: the per-token reduction in target perplexity produced by extending the revealed context past distance d, minus the corresponding marginal computed with the entire revealed span in scrambled order (Methods). Subtracting the shuffled-token baseline controls the predictive benefit available from the same token composition in scrambled order, so $P(d)$ isolates the additional contribution of intact sequential arrangement, resolved as a function of distance.

Figure 1 shows $P(d)$ for each of the ten corpora, plotted against distance on log–log axes. Across all corpora, $P(d)$ is positive throughout the measured range: at distances approaching 1000 tokens (the far end of our measurement range), prior context continues to reduce target perplexity by a measurable margin relative to shuffled controls. Long-range contextual influence is not confined to a recency window but extends as a graded function across the full sequences we examined.

The analyses that follow characterize this function: its form, a power law with exponent near 1 that recurs across languages, genres, and modalities; its source, sequential structure in language itself rather than properties of the probe, established by synthetic, cross-probe, and cross-domain controls; and its composition, the separable contributions of sentence-internal order, discourse sequence, and sentence content, distributed broadly across prior context rather than concentrated in privileged sentences.

2.2 Heavy-tailed decay across languages, genres, and modalities

The functional form of $P(d)$ is well approximated by a heavy-tailed decay. Across languages, genres, and modalities, $P(d)$ decays smoothly across nearly two orders of magnitude with a mean decay exponent $\alpha = 1.04$ (SD = 0.15; $P(d) \propto d^{-\alpha}$) — close to a $1/d$ power law — and a median goodness-of-fit of $r^2 = 0.96$. A reduced-sample replication averaging 20 shuffled permutations per target produced the same cross-corpus mean exponent ($\alpha = 1.04$) and improved fit stability (Supplementary Information), indicating that the headline result does not depend on the selected

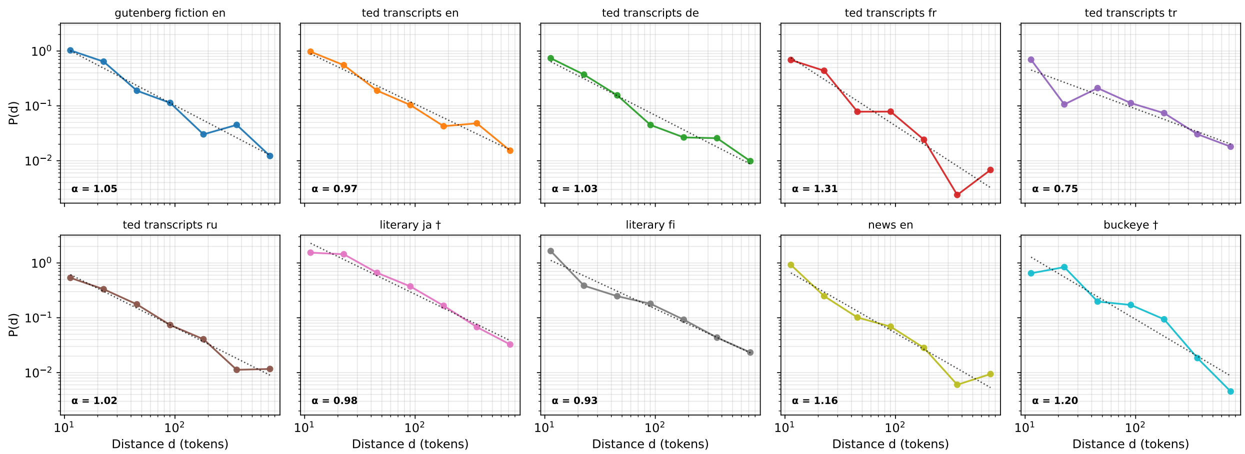

Figure 1: Cross-corpus contextual persistence functions. Per-token order-specific contextual influence $P(d)$ as a function of distance d, log–log axes. Each panel is one of ten corpora spanning six language families (Germanic, Romance, Turkic, Slavic, Japonic, Uralic) and four production modes. All cells exhibit heavy-tailed decay across nearly two orders of magnitude in distance, with mean decay exponent $\alpha = 1.04$ (SD $= 0.15$; $P(d) \propto d^{-\alpha}$, so the log–log slope is $-\alpha$) — close to a $1/d$ law. Points are shown for $d \geq 10$ tokens; the dotted line is the power-law fit. Intervals below $d = 10$ are excluded because at such short context the shuffled-token baseline is unstable: with only one or two tokens there is no meaningful permutation, so the order-specific difference is ill-defined there and does not reflect the persistence decay (Methods). The two cells marked † (Japanese literary prose; Buckeye spontaneous English speech) show mild downward curvature and are marginally better fit by a log-normal and a stretched-exponential form respectively, but remain heavy-tailed with the power law a close second.

permutation. Fits are computed on $d \geq 10$ tokens, the range over which both ordered- and shuffled-context bases are stable; very-short-range intervals are reported in Methods. In a comparison of five candidate functional forms by small-sample-corrected information criteria (Supplementary Information), the power law is preferred in eight of the ten corpora and outperforms a pure exponential in all ten; a fixed-scale exponential is never preferred. The two remaining corpora, Japanese literary prose and Buckeye spoken English, show mild downward curvature and are marginally better described by a log-normal and a stretched-exponential form respectively; in both the decay remains heavy-tailed and extends well beyond the range expected from a fixed temporal buffer acting alone.

The cross-corpus consistency of the decay regime is notable. The sample spans Germanic, Romance, Turkic, Japonic, Uralic, and Slavic language families, as well as fiction and literary prose, news and expository prose, prepared speech transcripts, and spontaneous speech. A narrow heavy-tailed regime across this range indicates that the CPF reflects a general property of natural language rather than a property of any single language, register, or modality.

The qualitative pattern that defines $P(d)$ — a positive contribution of ordered context at long range together with a near-zero contribution of shuffled context — extends beyond the six families covered by the long-range run. In a complementary cross-language analysis evaluating shorter context lengths on both probe models, the long-range sign pattern is preserved across three additional language families: Afro-Asiatic, Koreanic, and Sino-Tibetan (Supplementary Information). The headline exponent of $P(d)$ is therefore obtained on a six-family set within a broader nine-family cross-language replication of the qualitative pattern that defines the persistence function.

2.3 An exponent near 1: uniform influence across logarithmic timescales

Two properties of the decay should be separated. The first is that it is scale-free: a power law has no distance at which its behaviour changes character, and our fits show none across the measured range. The second, and more distinctive, is the value of the exponent. Across the ten corpora the fitted decay exponent is close to 1 and clustered around it (mean $\alpha = 1.04$, SD $= 0.15$, with $P(d) \propto d^{-\alpha}$). The observed between-corpus spread modestly exceeds document-sampling error (typical bootstrap SE ≈ 0.10; Supplementary Information), although permutation sensitivity analyses indicate that part of this spread reflects noise in the shuffled baseline. The exponent is therefore best described as clustered near 1, with limited evidence for residual corpus-level heterogeneity. The reproducibility of the near-unity value across otherwise very different samples is itself part of the result, and it carries a specific quantitative meaning.

For a CPF of the form $P(d) \propto d^{-\alpha}$, the aggregate contextual influence contained within a logarithmic interval of distance — the band spanning one multiplicative step, or one decade, of the past — is proportional to $d\,P(d) \propto d^{1-\alpha}$. When $\alpha \approx 1$ this quantity is approximately constant. Put concretely, if one divides the past into logarithmic bands (1–10, 10–100, and 100–1,000 tokens back), each band contributes approximately the same aggregate contextual influence, even though the influence of any single word continues to decline with distance. The reason the decades flatten is a near-exact cancellation of two opposing effects. Each successive logarithmic band reaches proportionally further back and so contains proportionally more words, and at $\alpha \approx 1$ that growth in number almost exactly offsets the $1/d$ decline in the average influence per token, so the many weakly contributing words far back combine to match the few strongly contributing words nearby. Contextual influence is therefore distributed approximately uniformly across logarithmic timescales: no logarithmic timescale dominates the aggregate, and every decade of the past contributes comparably to ongoing prediction (Figure 2).

This should not be read as memory being equally strong at all distances; individual words do become progressively less influential as they recede. What is approximately equal is the *aggregate* influence summed within each logarithmic decade: pointwise influence falls as $1/d$, while cumulative influence per multiplicative interval of distance remains flat.

The value $\alpha \approx 1$ is also a boundary. For $\alpha > 1$, aggregate influence is concentrated near the present and the remote past contributes negligibly in sum; for $\alpha < 1$, the sum is increasingly dominated by distant context. $\alpha \approx 1$ is the crossover between these regimes: the exponent at which no timescale dominates the total. That human language sits at this boundary, rather than at an arbitrary point along the continuum, is what most distinguishes the result from a generic heavy tail.

The near-unity regime is robust to the choice of response scale: expressing influence in additive log-probability (nats) units rather than perplexity leaves every corpus a near-$1/d$ power law, though the mean exponent shifts modestly from $\alpha = 1.04$ to $\alpha = 1.18$ (Supplementary Information), indicating approximate rather than exact equality of influence across logarithmic decades.

The reproducibility of the near-unity exponent across corpora is what supports treating this as a candidate scaling law rather than an isolated curve fit. As with Zipf's law, the recurrence of the value — an exponent near 1 across languages, modalities, and independently trained probe models — is the phenomenon, not merely the power-law form.

2.4 The CPF reflects sequential structure, not properties of the probe

A central concern in any LLM-based measurement is whether the observed pattern reflects properties of natural language or properties of the probe model. We address this directly with synthetic-

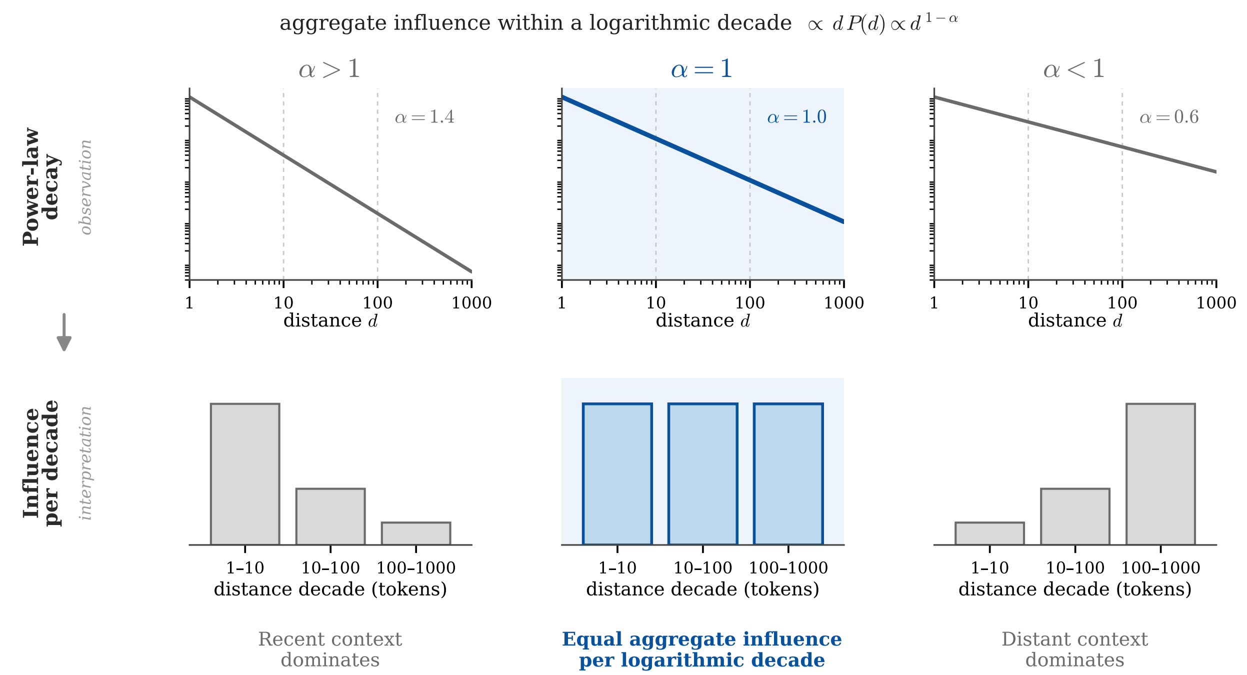

Figure 2: Persistence functions $P(d) \propto d^{-\alpha}$ at three exponents and the resulting per-decade influence (schematic). Top row (observation): the persistence law $P(d) \propto d^{-\alpha}$ on log–log axes, where the decay appears as a straight line of slope $-\alpha$. The three power laws look nearly indistinguishable. **Bottom row (interpretation):** their sharply different consequences. The aggregate influence within a logarithmic decade of distance is proportional to $d\,P(d) \propto d^{1-\alpha}$, so for $\alpha > 1$ (left) aggregate influence is concentrated in recent context; for $\alpha < 1$ (right) it is dominated by distant context; and at $\alpha \approx 1$ (centre) each logarithmic decade of the past — 1–10, 10–100, 100–1,000 tokens back — contributes approximately equally, distributing influence uniformly across logarithmic timescales. The measured mean exponent ($\alpha = 1.04$) places human language at this boundary. The lower bands are a conceptual illustration, not data.

sequence controls. We constructed sequences with token-frequency statistics matched to natural text but with sequential structure removed. Raw perplexity over these synthetic sequences exhibits apparent power-law structure with a slope comparable in magnitude to that observed on natural text (Llama-3.1: ≈ -1.27 on uniform random vocabulary; Mistral-7B: ≈ -1.34), replicating the general tendency for language-model perplexity to fall as context length increases even when genuine long-range dependence is absent.

The order-specific CPF, $P(d)$, however, is near zero across all distances on the synthetic sequences. The shuffled-token baseline that defines $P(d)$ absorbs the perplexity reduction attributable to token statistics alone, isolating the contribution of sequential organization. The long-range, heavy-tailed $P(d)$ observed for natural language therefore depends on properties of the linguistic sequence itself, not on the probe. We replicated the persistence signature with two independently trained probe models, Llama-3.1-8B and Mistral-7B, obtaining the same qualitative pattern and comparable decay ranges on shared corpora, along with near-zero $P(d)$ on synthetic controls. The CPF is therefore not an idiosyncratic property of the model used to estimate it.

A stronger version of this argument replaces synthetic sequences with real non-language sequences that carry their own long-range structure, each probed with its own domain-native autoregressive model. If the law were a by-product of autoregressive probing rather than a property of language, the identical CPF protocol should reproduce it elsewhere. It does not (Figure 3). On

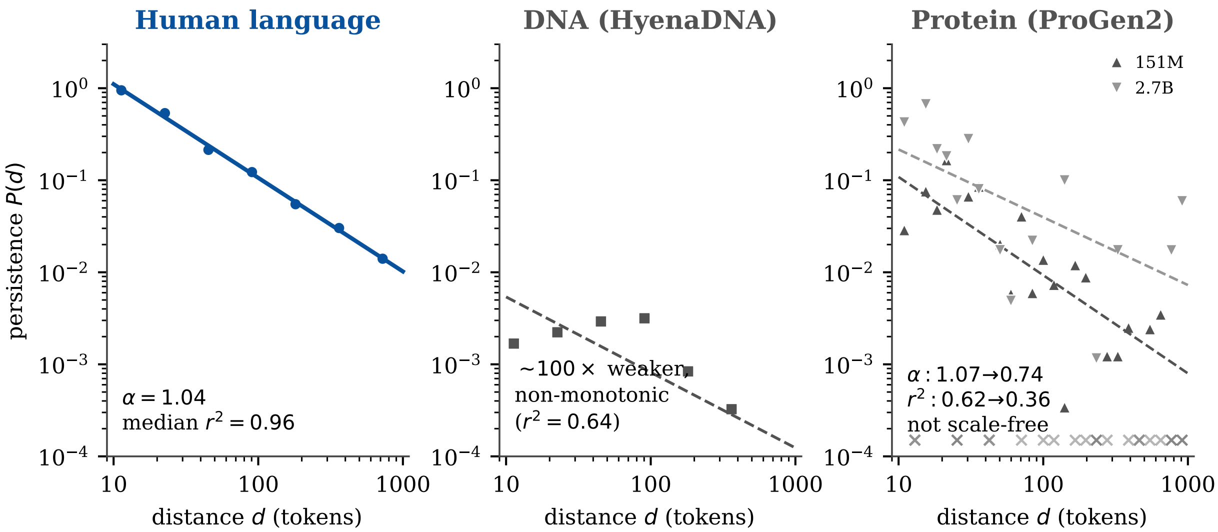

Figure 3: Order-specific persistence $P(d)$ for human language, DNA, and protein under the identical protocol. Order-specific persistence $P(d)$ under the identical CPF protocol, on log–log axes, for human language (aggregate of the ten corpora), DNA (HyenaDNA), and protein (ProGen2 at two scales, 151M and 2.7B). **Left:** language decays as a clean power law (mean $\alpha = 1.04$, median $r^2 = 0.96$; points lie on the fit). **Centre:** DNA is roughly two orders of magnitude weaker and non-monotonic, with no clean scale-free decay. **Right:** protein carries genuine long-range influence but is not scale-free — the fit is poor and the point estimate varies across model scale while power-law fit quality declines ($\alpha = 1.07 \rightarrow 0.74$, $r^2 = 0.62 \rightarrow 0.36$ from 151M to 2.7B); $\times$ marks distances where $P(d) \leq 0$. The discriminating combination is scale-freeness and exponent stability, not the mere presence of long-range structure.

genomic sequence (human chromosome 21, HyenaDNA), $P(d)$ is near zero, non-monotonic, and far weaker than language, with no scale-free decay. On protein sequence (Swiss-Prot, ProGen2), there is genuine long-range influence (expected from coevolutionary structure) but not language's form: the power-law fit is poor, and the point-estimate exponent varies across model scale while the fit itself deteriorates ($r^2 = 0.36$–0.62 versus 0.96 for language; Supplementary Information), so the curve is long-range but not scale-free. Long-range dependence is not unique to language. The distinguishing result is the combination of scale-free decay, a near-unity exponent, and cross-probe consistency: among the domains and probes tested, only human language shows all three, with the measurement protocol and the presence of long-range structure held in common. The regularity is a property of the linguistic sequence, not of the instrument.

2.5 Sequence, reversal, and content make distinguishable contributions to persistence

To separate the effects of disrupting, reversing, and dissolving discourse structure, we evaluated four context conditions on the same target windows (Figure 4): intact context; sentence-shuffled context, which preserves sentence content and within-sentence order while disrupting the discourse sequence; sentence-reversed context, which preserves the set of sentence adjacencies while reversing their order and target-relative positions; and token-shuffled context, which removes ordered structure altogether. Three contrasts follow. The sentence-reversal contrast (intact minus sentence-reversed) measures the loss associated with reversing the temporal organization of the prior discourse, while preserving sentence-internal order and the set of sentences; the sequence-permutation contrast (intact minus sentence-shuffled) isolates sentence-sequence organization; and the sentence-content contrast (sentence-shuffled minus token-shuffled) estimates the contribution retained by sentence content once discourse ordering is removed.

Over their positive range, each contrast is approximately linear on log–log axes, with distinct fitted slopes. The sentence-reversal and sequence-permutation contrasts are steep and concentrated at short range (slopes -1.44 ± 0.48 and -1.26 ± 0.01 respectively, approaching zero by $d > 100$ tokens), whereas the sentence-content contrast is shallow and carries the long-range tail (slope -0.78 ± 0.24). Sensitivity to sentence reversal and discourse-sequence permutation is therefore concentrated at short range, whereas sentence-level content carries the long-range tail.

A localized order manipulation converges on the same decomposition. As an independent check, we applied an increment-shuffle manipulation that localizes the disruption to a single distance band: for each adjacent pair of context lengths (c_{i-1}, c_i), all nearer context (distances $1 \ldots c_{i-1}$) is held intact and in place, and only the tokens at distances $c_{i-1} + 1 \ldots c_i$ are permuted within that band ($K = 20$ permutations per target and band). The per-token quantity $Q(d) = (\overline{\text{ppl}}_{\text{shuf}} - \text{ppl}_{\text{intact}})/(c_i - c_{i-1})$, at the geometric midpoint $d = \sqrt{c_{i-1} c_i}$, isolates the internal order of a single distant span while the surrounding context is preserved; where $P(d)$ reflects the order-specific value of the accumulating context as a whole, $Q(d)$ probes span-internal order alone. Across the eight cells with a range-matched $P(d)$ estimate (Figure S1), $Q(d)$ is positive throughout but decays more steeply than $P(d)$, with a log–log slope near -1.3 and downward curvature (bounded and curved forms outperform a pure power law by AIC in most cells), whereas the range-matched $P(d)$ retains a slope near -1.0; the band-local exponent exceeds the integrated exponent in every cell tested. This coincides with the sequence-permutation contrast (-1.26 ± 0.01) recovered from sentence shuffling, obtained here by an independent manipulation at token resolution with nearer context held fixed. Two structure-disrupting methods therefore converge: the internal ordering of localized spans makes a steep, short-range, bounded contribution, while the scale-free long-range tail is carried by integrated content rather than by the internal order of any distant region. We note what this contrast does and does not license. In natural generation no span is conditioned on in isolation; each word is produced against the entire ordered sequence so far. The extension-marginal $P(d)$ therefore tracks a quantity with a direct referent in natural generation — the influence that accrues as ordered context lengthens — whereas the increment shuffle is a decomposition tool, informative precisely because it shows where the scale-free law does not originate. The persistence law is a property of integrated ordered context, not a sum of separable span contributions: distant material exerts its influence as part of the ordered whole rather than through its internal arrangement alone.

2.6 Persistence is broadly distributed across prior sentences

We next asked whether persistence is concentrated in a small number of privileged prior sentences or broadly distributed across the prior context. We addressed this with a sentence-level ablation: for each target, we removed one prior sentence at a time and measured the resulting change in prediction.

The majority of prior sentences contribute positively to prediction at all distances examined. The proportion of contributing sentences declines gradually with distance, from approximately 90% at short range to 55–60% beyond 500 tokens, with no sharp transition or threshold (Figure 5). The magnitude of contribution is concentrated at recent positions (top-5 sentences account for 30–51% of total positive influence; Gini coefficient 0.78–0.90), but this concentration is attributable to recency. After matching for distance exactly, influence by position is flat in TED transcripts; fiction

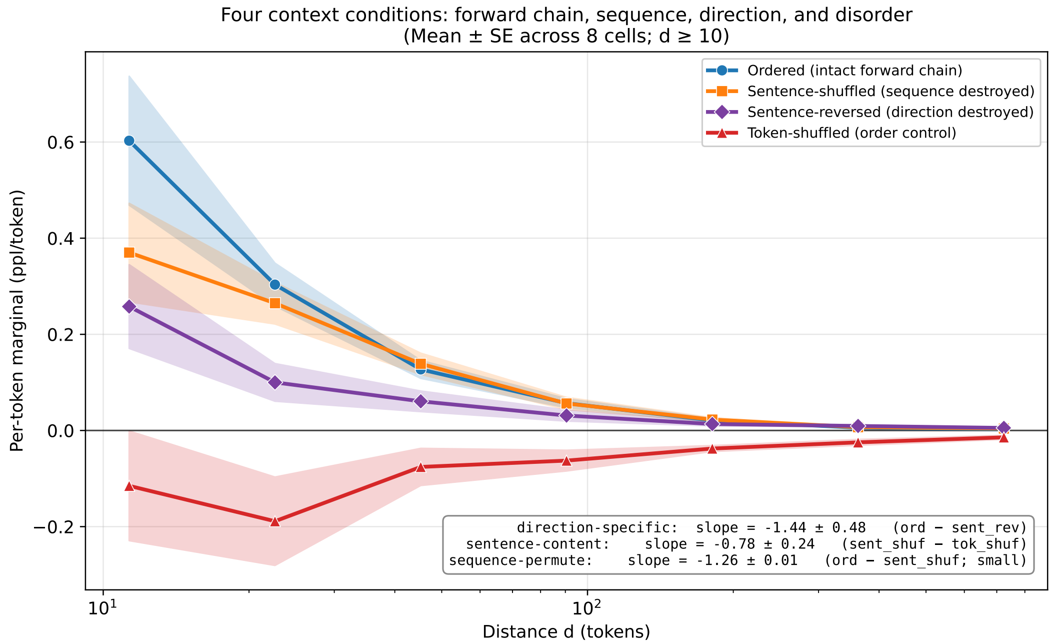

Figure 4: Per-token order-specific marginals under four context conditions. Per-token order-specific marginals for four context conditions on log–log axes, aggregated across eight cells: ordered (intact forward chain), sentence-shuffled (discourse sequence destroyed, sentence content preserved), sentence-reversed (sentence order reversed, so the set of adjacencies is preserved while their order and target-relative positions are reversed), and token-shuffled (ordered structure removed). Curve ribbons are mean ± SE across the eight cells. Three contrasts decompose $P(d)$: sentence-reversal (intact minus sentence-reversed; slope -1.44 ± 0.48), sequence-permutation (intact minus sentence-shuffled; slope -1.26 ± 0.01), and sentence-content (sentence-shuffled minus token-shuffled; slope -0.78 ± 0.24); slope uncertainties are the SD of the per-cell slopes. The sentence-reversal and sequence-permutation contrasts are concentrated below $\sim$100 tokens, whereas the sentence-content contrast is shallow and carries the long-range tail. The reported quantities are log–log slopes, not values of α; since $P(d) \propto d^{-\alpha}$ with $\alpha > 0$, a slope of -1.44 corresponds to a decay exponent $\alpha = 1.44$.

shows one small, interpretable exception — a $\sim$0.05 elevation at the earliest one to two positions, an order of magnitude below mean per-sentence influence, consistent with the cost of removing sentences that introduce recurring referents (Figure 5b). Persistence is therefore a distributed field of influence across the prior sequence, with magnitude graded smoothly by recency rather than localized to a small number of anchoring positions.

3 Discussion

Our observations establish a scaling law, which we term the contextual persistence law: contextual influence in human language decays as an approximately scale-free power law, with a near-constant exponent across ten corpora, six language families, and both written and spoken modalities. Because that exponent is close to 1, contextual influence is distributed approximately uniformly across logarithmic timescales rather than concentrated at any one. Unlike generic heavy-tailed behaviour, an exponent near unity occupies the unique boundary at which aggregate influence per logarithmic decade is approximately constant — for $\alpha > 1$ influence concentrates at short range, for $\alpha < 1$ at long range — so the law specifies not merely the absence of a characteristic scale but a particular

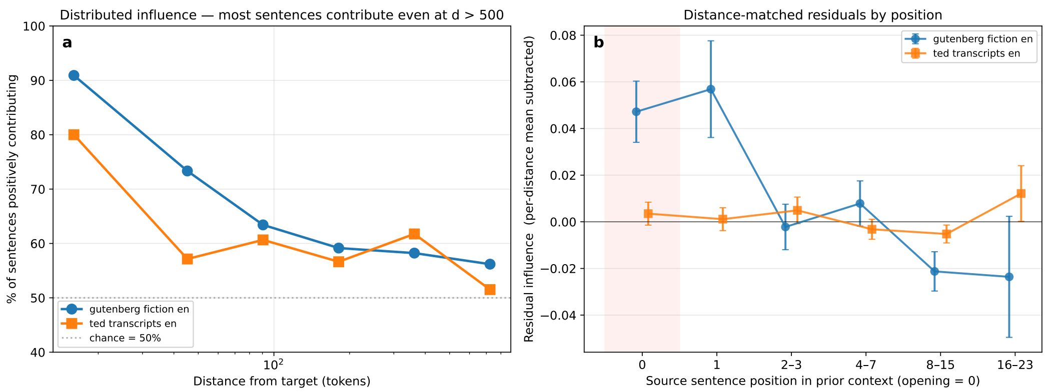

Figure 5: Persistence is broadly distributed across prior sentences. a, Proportion of prior sentences whose removal degrades target prediction, as a function of distance to the target. The proportion declines smoothly from ∼90% within the first 32 tokens to ∼55–60% beyond 500 tokens, with no sharp transition. The top-5 sentences account for 30–51% of total positive influence (Gini 0.78–0.90), but this concentration reflects recency, not a privileged structural role for any specific position. **b,** Mean per-sentence influence by absolute position in the prior context, after subtracting the mean influence at each exact sentence distance (nonparametric distance control; leave-one-sentence-out influence matrices, 15 documents per corpus). Positions 0 and 1 are shown individually; later positions are aggregated in dyadic bins (2–3, 4–7, 8–15, 16–23). Shaded band marks the opening position; error bars, s.e.m. TED residuals are flat across positions. Fiction shows a small elevation at the earliest two positions (∼0.05, an order of magnitude below its mean per-sentence influence), consistent with the added cost of removing sentences that introduce recurring referents. No position carries influence beyond distance on a scale comparable to the distance effect itself: influence is a distributed field graded by recency, not anchored at privileged positions.

organization of influence across scales. The consistency of the exponent across such diverse samples makes this a candidate cross-linguistic regularity of human language, in the same empirical sense that Zipf's and Heaps' laws are cross-linguistic regularities. In the hierarchy of statistical descriptions considered here — unit frequency (Zipf's law), pairwise dependence across distance (long-range mutual information), and order-specific dependence — the contextual persistence law is the third, order-specific rung. Where previous scaling laws describe distributions of units or pairwise relations among them, this law characterizes the scaling of order-specific contextual influence: the sequential organization of predictive dependence across distance, the component that remains once unit identity and co-occurrence are removed. It is a law to be explained rather than an adjudication between mechanisms: any account of how human language is generated must explain why its statistical product lacks a detectable characteristic scale over this range. An architecture with a fixed maintenance horizon would generally be expected to produce an inflection in $P(d)$ near that horizon, of whatever size; we observe none across nearly two orders of magnitude. An account in which long-range coherence rests on a small number of privileged anchor positions would generally predict a concentrated rather than a distributed influence field; we observe broad, recency-graded distribution. And an account in which distant material contributes through the internal structure of isolated episodes would predict that the law could be reconstructed from span-internal order; the increment-shuffle experiment shows it cannot: band-local order decays too steeply, and the scale-free law emerges only at the level of integrated ordered context. We do not take these observations to settle how memory is organized during production; they characterize the product, not the process. But the law they describe is what every proposed process must reproduce.

Why should such a law exist? The mean exponent we report ($\alpha = 1.04$, SD $= 0.15$) sits in a neighborhood independently identified in memory research by methods that involve no language model: the odds that a memory is needed in natural environments [Anderson and Schooler, 1991], associative decay in free recall [Kahana et al., 2002], and the form of forgetting across human and animal tasks [Wixted and Ebbesen, 1991] all follow power functions with order-unity exponents. We do not claim a single mechanism generates them all, but the recurrence raises the possibility (a hypothesis, not a finding) that the persistence law reflects the same constraints operating in human language generation: the accessibility of prior content during production falling off in the same graded, scale-free way that memory itself decays. If this account is accurate, it dissolves a familiar discontinuity. Architectures that divide memory into a capacity-limited store for the present and a separate long-term store for the past should leave some boundary signature in the product, and none appears: the law is continuous across the range where any such boundary would fall, consistent with graded rather than architecturally bounded characterizations of memory [Cowan, 2001, Ericsson and Kintsch, 1995, Ma et al., 2014]. The observation doing the work remains the same: a near-$1/d$ persistence law whose near-unity exponent places language at the boundary where each logarithmic timescale contributes approximately equally to prediction.

The CPF characterizes the statistical structure of language, not directly the memory process that produced it: it is computed over text, and predictive structure in text need not imply that speakers and writers actively used long-range memory to generate it. The property is nonetheless far more constrained than a generic statistical regularity: because $P(d)$ subtracts a shuffled-token baseline, vanishes on token-scrambled and frequency-matched synthetic sequences, and replicates across independently trained probes, the long-range dependence reflects genuine order-dependent structure in what humans produce, not lexical co-occurrence or measurement bias. A related objection, that written text is a static visual trace an author can edit and cross-reference offline unlike online cognition, is tempered by the modalities in which the law holds: the same signature appears in spontaneous conversational speech, where offline revision is impossible, and in the spoken delivery of prepared TED discourse. The observation across both spontaneous and prepared speech indicates that the law is not contingent on written revision.

A parsimonious reading of these results is that the human generative process is itself autoregressive: each word is produced against the accumulated ordered history of the sequence, and the persistence law is the trace that this mechanism leaves in its product. On this reading the probe is not an arbitrary instrument but one matched in form to the process it measures. We advance this as an interpretation, not a demonstrated mechanism: the measurements characterize the product, and other generative architectures — hierarchical planning, retrieval over structured discourse representations — could conceivably yield the same scale-free profile. Nor can modern language models decide the question: they are explicitly autoregressive, but they are trained on human text and inherit its statistics, so their reproduction of the law shows only that they have learned it. What would adjudicate is evidence from the human process itself. Converting the present result from a property of the product into a claim about the producer, or the comprehender, will require online behavioural and neural evidence, and the law makes this directly testable because it predicts a specific quantity. The central prediction is that the cost of disrupting prior context should itself scale as $P(d)$: shuffling or replacing context in log-spaced windows (for example, tokens 10–20 back versus 100–110 back) should slow human reading and lengthen gaze durations on a downstream target [Hale, 2001, Levy, 2008, Smith and Levy, 2013, Futrell et al., 2020] by amounts that fall off with distance at the near-unity rate, rather than dropping sharply at a working-memory horizon. Parallel predictions follow for proactive interference during comprehension and for the context-driven reduction of the N400 [Kutas and Federmeier, 2011], each expected to follow the same $\alpha \approx 1$ decay; we regard adjudicating these as the key open hypothesis the law motivates.

Human language exhibits a simple quantitative law, the contextual persistence law, in which the influence of prior context is distributed approximately uniformly across logarithmic distances in human-produced language. Whether this regularity reflects properties of memory, discourse organization, planning, prediction, or communication is precisely the mechanistic question the law leaves open; whether it generalizes to non-linguistic sequential behaviours with extended temporal coherence (music improvisation, motor sequences, drawing) remains to be tested. But like other scaling laws in language, its value lies less in any mechanism it immediately reveals than in the constraint it imposes on every mechanism proposed thereafter: its existence is now an empirical target that any theory of language must account for.

4 Methods

4.1 Corpora

The long-range CPF was computed on ten corpora: English fiction (public-domain texts from Project Gutenberg, `https://www.gutenberg.org`), English news prose (the English/BBC subset of XL-Sum [Hasan et al., 2021]), English spontaneous speech (Buckeye Corpus of Conversational Speech [Pitt et al., 2007]), TED transcripts in English, German, French, Turkish, and Russian (OPUS TED2020 [Reimers and Gurevych, 2020, Tiedemann, 2012]), and literary prose in Japanese (Aozora Bunko) and Finnish (Project Gutenberg). Together these cover six language families (Germanic, Romance, Turkic, Slavic, Japonic, Uralic) and four production modes (written fiction, news prose, prepared speech, spontaneous speech). Per-corpus document counts (documents contributing at least one full-length target window) were: English fiction 60, English news 60, Buckeye 26, TED English 60, German 60, French 31, Turkish 12, Russian 60, Japanese literary 42, Finnish literary 38 — 449 documents in total. The complementary cross-language qualitative replication of the sign-flip pattern (Afro-Asiatic, Koreanic, Sino-Tibetan) was performed on Wikipedia and TED-transcript cells from a broader corpus inventory at shorter context lengths. Full per-corpus provenance — sources, identifiers, licenses, snapshot dates, and selection criteria — is given in Supplementary Information (Table S3), and preprocessing steps are described there.

4.2 Probe models

Two autoregressive language models served as probes: Llama-3.1-8B (base; `Meta-Llama-3.1-8B`) [Grattafiori et al., 2024] and Mistral-7B-v0.1 (`mistralai/Mistral-7B-v0.1`) [Jiang et al., 2023]. Neither was fine-tuned for this study. The probes were used only to compute conditional probability estimates over fixed token sequences; the language models themselves were not the object of study. Headline results were obtained on Llama-3.1-8B; the qualitative pattern was replicated on Mistral-7B for shared corpora as a probe-independence check.

4.3 Contextual persistence function

For each document, a single target region of 30 tokens was selected at the midpoint (50%) of document length. Context preceding the target was revealed in incrementally larger spans at log-spaced context lengths $c \in \{0, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024\}$. At each context length, target perplexity was computed under intact context (the actual preceding c tokens, original order) and shuffled context (the same c tokens, random order). The shuffled baseline used a single random permutation of the prior-context tokens at each target and context length, drawn with a fixed seed so that the baseline is deterministic and reproducible and the same permutation is applied across probe models; the 30-token target itself was never permuted, and no sentence-boundary or special tokens were introduced into the shuffled span. Sensitivity of the fitted exponent to this single-permutation choice is quantified in Supplementary Information (*Robustness of the exponent: sensitivity to the shuffled permutation*), where the analysis is repeated with $K = 20$ independent permutations per target and context length. Per-distance marginals were obtained as the per-token change in perplexity across successive context lengths, $m_d = (\mathrm{ppl}_{c_{i-1}} - \mathrm{ppl}_{c_i})/(c_i - c_{i-1})$, with distance indexed by the geometric mean $d = \sqrt{c_{i-1}c_i}$ of the interval. The CPF was defined as $P(d) = m_d^{\text{intact}} - m_d^{\text{shuffled}}$. The shuffle preserves the identity and composition of the prior context while destroying its order, so this subtraction separates the predictive benefit of intact order from the benefit available from the same token composition in scrambled order, and from probe-internal token-frequency calibration. Marginals were aggregated across documents within each corpus and binned by distance. Only documents with at least $1024 + 30 + 50$ tokens were included, ensuring a full-length context window and target for every measurement.

4.4 Functional-form fitting

The per-token order-specific gap was fit to five candidate functional forms (power law, exponential, stretched exponential, truncated power law, and quadratic-in-log, i.e. log-normal-in-distance) and ranked per corpus by the small-sample-corrected Akaike criterion (AICc); the full comparison is reported in Supplementary Information. Fits were performed in log space on the fit range $d \geq 10$ tokens, the first interval at which both ordered and shuffled bases are at least eight tokens; shorter intervals were excluded a priori because with only one or two context tokens there is no meaningful permutation, leaving the shuffled baseline, and hence $P(d)$, unstable at $d < 10$. Within this fixed fit range the aggregated (corpus-mean) $P(d)$ is positive at every distance for all ten language corpora, so no bins were dropped from any language fit; the positive-value requirement is therefore not a source of tail-selection bias for the headline result and becomes operative only for the non-language controls, where sign changes occur and we report the number of positive bins used (Figure 3).

4.5 Synthetic-sequence controls

To distinguish probe-internal calibration dynamics from genuine sequential dependence, two synthetic-sequence controls were constructed: uniform-vocabulary sequences (tokens drawn uniformly at random from the model vocabulary) and frequency-matched sequences (tokens drawn with empirical token-frequency probabilities matched to the natural corpora). The same CPF protocol was applied to both.

4.6 Non-language domain controls

To test whether the persistence law is specific to human language or a generic property of autoregressive probing over any structured sequence, the identical CPF protocol was applied to two non-language biological-sequence domains, each with a domain-native autoregressive probe. For DNA, the probe was HyenaDNA [Nguyen et al., 2023], a single-nucleotide autoregressive genomic language model, run at two scales (`hyenadna-medium-160k-seqlen` and `hyenadna-large-1m-seqlen`); sequences were non-overlapping ACGT windows from human reference chromosome 21 (GRCh38, Ensembl [Cunningham et al., 2022]). For protein, the probe was ProGen2 [Nijkamp et al., 2023], an autoregressive decoder protein language model, run at two scales (`progen2-small`, 151M; `progen2-large`, 2.7B); sequences were UniProtKB/Swiss-Prot proteins [The UniProt Consortium, 2023] restricted to the twenty standard amino acids. In both domains the protocol matched the language runs: a 30-residue target at the sequence midpoint, log-spaced context lengths capped at the model's context window, an intact-versus-shuffled contrast, and power-law fitting on positive bins at $d \geq 10$. Because these ports are custom, probe faithfulness was verified before analysis:

held-out per-residue perplexity was reproduced (ProGen2: 15.1 and 15.7 for the small and large models respectively, with per-protein spread and well below the ≈ 20 uniform-amino-acid baseline), and a short-range order-specific floor was confirmed. For protein, the exponent was estimated on a dense log-spaced grid (approximately 28 bins between $d = 10$ and 993) over 300 documents at each scale, with 95% confidence intervals from a 1,000-fold document bootstrap using the same estimator applied to the language corpora. Full domain-control results and diagnostics are reported in Supplementary Information.

4.7 Sentence-level decomposition

The effects of sentence-order reversal, discourse-sequence permutation, and within-sentence content were assessed by computing $P(d)$ under four context conditions on the same target windows: ordered (intact); sentence-shuffled (the prior context segmented at sentence boundaries and the sentences randomly permuted, with within-sentence token order preserved); sentence-reversed (the order of the sentences reversed, so that the sentence nearest the target is moved farthest from it, with within-sentence token order preserved), which preserves the set of sentence adjacencies while reversing their order and target-relative positions; and token-shuffled (all context tokens randomly permuted). From these we formed three contrasts, each fit by least squares on log–log axes over the positive bins at $d \geq 10$: the sentence-reversal contrast (ordered minus sentence-reversed), which measures the loss from reversing the temporal organization of the prior discourse; the sequence-permutation contrast (ordered minus sentence-shuffled), which isolates sentence-sequence organization; and the sentence-content contrast (sentence-shuffled minus token-shuffled), which estimates the contribution retained by sentence content. Slopes are reported as per-cell means $\pm$ SD across cells. The decomposition was computed across eight cells: the long-range corpora excluding Buckeye (spontaneous speech lacks unambiguous sentence boundaries) and Russian (added to the headline analysis after the decomposition run). Figure 4 aggregates these eight cells.

4.8 Sentence-level ablation

For documents in two corpora (English fiction and English TED transcripts, 20 documents per corpus, 1024-token prior context), each prior sentence was individually removed and the resulting change in target perplexity was computed (leave-one-sentence-out). Per-sentence influence was characterized by sign (positive = removing the sentence increased target perplexity) and magnitude. The proportion of contributing sentences was computed in distance bins, and concentration metrics (Gini coefficient over per-sentence magnitudes; share of total positive influence in the top-5 sentences) were computed per document and aggregated. To test whether magnitude variation across prior-sentence positions reflects a privileged structural role over and above distance from the target, the full leave-one-sentence-out influence matrix was computed on a 15-document subset of each ablation corpus (standardized 25-sentence prior-context windows). For the positional analysis, per-sentence influence values were demeaned within each exact sentence distance — a nonparametric distance control that imposes no functional form and requires no exclusions — and mean residuals were examined by absolute position, with positions 0 and 1 reported individually and later positions aggregated in dyadic bins (2–3, 4–7, 8–15, 16–23) for stable estimates. Distance-matched residuals are flat across positions in TED transcripts; in fiction the earliest one to two positions show a small positive residual ($\sim$0.05 per sentence, an order of magnitude below mean per-sentence influence), consistent with the reintroduction cost of referents established at the opening. No structural position carries influence beyond distance on the scale of the distance effect itself.

Supplementary Information: functional-form robustness

Because scale-free claims are sensitive to the choice of fitting procedure and to competing functional forms, we treated functional-form selection as a robustness analysis rather than assuming a power law. For each corpus the per-token order-specific gap was fit, in log space, to five candidate forms: a power law ($\log y = a + b \log d$), an exponential ($\log y = a + bd$), a stretched exponential ($\log y = a - \beta d^{\gamma}$), a truncated power law ($\log y = a + b \log d - d/\lambda$), and a quadratic-in-log (log-normal-in-distance) form ($\log y = a + b \log d + c(\log d)^2$). Models were ranked per corpus by the small-sample corrected Akaike criterion (AICc), which penalizes the additional free parameters of the more flexible forms; each fit used the seven binned points with $d \geq 10$ and positive influence.

The power law is preferred in eight of the ten corpora (Table S1) and outperforms the pure exponential (the canonical fixed-scale alternative) in all ten, by ΔAICc ranging from +0.3 to +16.8 (median $\approx +14$); the exponential is never preferred. The truncated power law never improves on the plain power law: in six corpora its fitted decay term is non-negative (no cutoff at all), and in three of the remaining four the implied cutoff λ lies beyond the measured range. Only Buckeye spontaneous speech shows a truncated-power-law cutoff within range ($\lambda \approx 276$ tokens). Separately, and as a distinct model comparison, Buckeye is also the one corpus for which the stretched exponential is marginally preferred over the power law by AICc, while Japanese literary prose marginally prefers the quadratic-in-log form. Both the cutoff and the stretched-exponential preference point to mild downward curvature at the longest distances in this cell, but they are separate tests and should not be read as one explaining the other. In both non-power-law cases the power law remains a close second (margin < 2 AICc), the fitted stretched exponent stays well below one, and the decay remains heavy-tailed. The fitted power-law exponents are tightly clustered near 1 (range $\alpha = 0.75$ to 1.31; eight of ten within 0.93 to 1.20). The maximum-likelihood, Kolmogorov–Smirnov procedure of Clauset et al. [2009] addresses whether a *sample* is drawn from a heavy-tailed distribution rather than the present setting of regressing a response, $P(d)$, on distance; we therefore compared candidate decay functions using AICc. No alternative form removes the central observation: across the measured range the decay carries no characteristic scale.

Corpus	N	Preferred form	ΔAICc (exp − power)	α [95% CI]
Gutenberg (fiction, en)	60	power law	+12.9	1.05 [0.90, 1.33]
TED (en)	60	power law	+15.5	0.97 [0.87, 1.11]
TED (de)	60	power law	+15.5	1.03 [0.93, 1.16]
TED (fr)	31	power law	+9.4	1.31 [1.02, 1.57]
TED (tr)	12	power law	+5.0	0.75 [0.37, 1.02]
TED (ru)	60	power law	+16.8	1.02 [0.94, 1.13]
Literary prose (ja)	42	quadratic-in-log	+13.2	0.98 [0.84, 1.15]
Literary prose (fi)	38	power law	+14.9	0.93 [0.77, 1.07]
News (en)	60	power law	+13.6	1.16 [0.98, 1.37]
Buckeye (speech, en)	26	stretched exp.	+0.3	1.20 [0.82, 1.53]

Table S1: Functional-form comparison and per-corpus exponents across the ten long-range corpora (Table S1). N is the number of contributing documents. ΔAICc (exp − power) is positive whenever the power law is preferred over a pure exponential; it is positive in every corpus, and the power law is the AICc-preferred form in eight of ten (the two exceptions show mild downward curvature, with the power law a close second). Exponent 95% confidence intervals are from a document-level bootstrap ($B = 2000$); the wider intervals correspond to the smaller corpora (Turkish, Buckeye, French).

Robustness of the exponent: units and heterogeneity

Two further checks bound the interpretation of the exponent. First, because $P(d)$ is defined as a difference of perplexities, and perplexity is exponential in the underlying log-loss, we recomputed the entire analysis with $P(d)$ defined in log-probability (nats) units (i.e., replacing each perplexity by its logarithm, an additive information quantity) and refit the power law. The scale-free *form* is robust to this choice: every corpus remains a near-$1/d$ power law and the ordering of corpora is preserved. The precise exponent, however, is not unit-invariant: the mean shifts from $\alpha = 1.04$ (perplexity) to $\alpha = 1.18$ (log-probability), a consistent offset of ≈ 0.14 present in every cell. We report perplexity-based values in the main text for continuity with the raw measurement and note that the order-unity exponent, and the associated approximate uniformity of influence across logarithmic timescales, hold in both unit conventions, with the exact value dependent on the choice.

Second, a document-level bootstrap ($B = 2000$ resamples of documents within each corpus) yields the per-corpus confidence intervals in Table S1. The between-corpus standard deviation of the ten point estimates (0.156) modestly exceeds the typical within-corpus bootstrap standard error (≈ 0.10). The permutation-sensitivity analysis below shows that part of this residual spread reflects noise in the single-permutation shuffled baseline rather than genuine linguistic variation: averaging over permutations narrows the apparent cross-corpus range. The exponent is therefore best described as clustered near 1, with limited evidence for residual corpus-level heterogeneity and the low-N corpora (Turkish, Buckeye, French) contributing most of both the apparent spread and the widest intervals.

Robustness of the exponent: sensitivity to the shuffled permutation

Because $P(d)$ subtracts a shuffled-context baseline, and the main analysis used a single random permutation of the prior context at each target and context length, we tested whether the fitted exponent depends on that choice with a permutation-seed sensitivity analysis. On a subset of up to 25 documents per corpus (all 26 for Buckeye), we redrew $K = 20$ independent permutations at each target and context length and recorded the target perplexity under every one, so that both the permutation-averaged baseline and the full across-permutation distribution of the exponent are available; permutation index 0 reused the original seed. Three results follow. First, the cross-corpus mean exponent is unchanged: averaging the shuffled baseline over the 20 permutations gives a mean $\alpha = 1.04$, and the mean of the 20 individual single-permutation estimates is $\alpha = 1.05$, both matching the single-permutation value of 1.04 in the main text. Second, averaging the baseline improves the power-law fits: corpora whose single-permutation fits were degraded by shuffle noise rise to $r^2 \approx 0.97$ once the averaged baseline removes per-permutation fluctuation in the shuffled perplexity (TED German $r^2\, 0.66 \to 0.97$; Finnish literary $0.55 \to 0.97$; TED French $0.82 \to 0.97$; TED Turkish $0.86 \to 0.98$). Third, the per-corpus exponent carries non-negligible permutation noise, concentrated in the smaller corpora: the across-permutation standard deviation of α reaches ≈ 0.20 (Finnish literary prose, English news) while the larger corpora vary by < 0.1, and a single permutation can differ from the 20-permutation average by up to ≈ 0.28 (TED French). Averaging accordingly narrows the apparent cross-corpus spread, indicating that part of the between-corpus heterogeneity in Table S1 reflects permutation noise rather than genuine linguistic variation. No corpus-level conclusion depends on the chosen permutation, and the near-unity, scale-free signature is unchanged. Per-corpus exponents under the single-permutation and $K = 20$ averaged baselines are given in Table S2.

Corpus	α (single perm.)	α (avg., $K = 20$)	r^2 (avg.)	SD_α (perms)
Gutenberg (fiction, en)	1.15	1.10	0.93	0.09
TED (en)	1.04	0.97	0.96	0.09
TED (de)	1.09	1.07	0.97	0.11
TED (fr)	1.43	1.16	0.97	0.12
TED (tr)	0.75	0.90	0.98	0.07
Literary prose (ja)	0.92	0.99	0.98	0.09
Literary prose (fi)	0.71	0.87	0.97	0.20
News (en)	1.22	1.19	0.95	0.18
Buckeye (speech, en)	1.20	1.11	0.95	0.16

Table S2: Permutation-seed sensitivity of the per-corpus decay exponent (Table S2), Llama-3.1-8B probe, on a subset of up to 25 documents per corpus (all 26 for Buckeye). "Single perm." reuses the original seed; "avg., $K = 20$" averages the shuffled baseline over twenty independent permutations per target and context length; SD_α is the standard deviation of the exponent across the twenty permutations. Averaging leaves the cross-corpus mean at $\alpha = 1.04$ while raising the poorer single-permutation fits to $r^2 \approx 0.97$. The Russian TED cell, computed with a separate pipeline in the main analysis, was not included in this check.

Corpus sources and provenance

Table S3 lists the source, license, and snapshot or version for each of the ten long-range corpora; preprocessing and selection are summarized here. English fiction comprises public-domain novels from Project Gutenberg (catalogue `pg_catalog.csv`; plain-text ebooks), selected as English fiction by Library of Congress class (PR, PS, PZ) or fiction subject headings, ordered by ascending Gutenberg ID to favour canonical works, with a per-author cap and rejection of verse, drama, and OCR artefacts, then segmented into ≈3,000-word chunks. English news is the English (BBC) portion of XL-Sum [Hasan et al., 2021]. English spontaneous speech is the Buckeye Corpus [Pitt et al., 2007, 2005], retaining interviewee turns. TED transcripts (English, German, French, Turkish, Russian) are from the OPUS TED2020 monolingual release [Reimers and Gurevych, 2020, Tiedemann, 2012] (version v1), retaining talks of at least 900 words after a per-language junk-ratio filter, one segment per talk. Japanese literary prose comprises public-domain texts from Aozora Bunko (Nippon Decimal Classification 913/914/915; 21 authors, 42 texts) and Finnish literary prose public-domain texts from Project Gutenberg (19 authors, 38 texts); both were captured in a snapshot dated 2026-05-01. The cross-language qualitative replication (Afro-Asiatic, Koreanic, Sino-Tibetan) drew additionally on Wikipedia (dumps current at the time of analysis; CC BY-SA 3.0), sampled at a 500-token minimum after removing markup, references, and infoboxes. Tokenization used each probe's native tokenizer throughout.

Data and code availability

Analysis code, corpus manifests, per-corpus persistence curves, and the figure-generation pipeline will be made publicly available in a repository with an assigned DOI upon publication; they are available from the author on reasonable request in the interim.

Author contributions

E.B. conceived and designed the study, performed the analyses, and wrote the manuscript.

Corpus	Source (reference)	License	Snapshot / version
English fiction	Project Gutenberg	Public domain (US)	2026 snapshot
English news	XL-Sum, BBC subset [Hasan et al., 2021]	See source	`csebuetnlp/xlsum`
English speech	Buckeye [Pitt et al., 2007, 2005]	Free for research	2007 release
TED (en, de, fr, tr, ru)	OPUS TED2020 [Reimers and Gurevych, 2020, Tiedemann, 2012]	See source	v1
Literary prose (ja)	Aozora Bunko	Public domain	2026-05-01
Literary prose (fi)	Project Gutenberg	Public domain (US)	2026-05-01
Wikipedia (replication)	Wikimedia dumps	CC BY-SA 3.0	Dump at analysis time

Table S3: Sources and provenance for the language corpora (Table S3). "See source" indicates the distributing repository states the terms of use; no separate license file was recorded at ingestion. Snapshot dates are given where recorded; the Project Gutenberg fiction set was captured in the same 2026 collection window but without a per-file date stamp.

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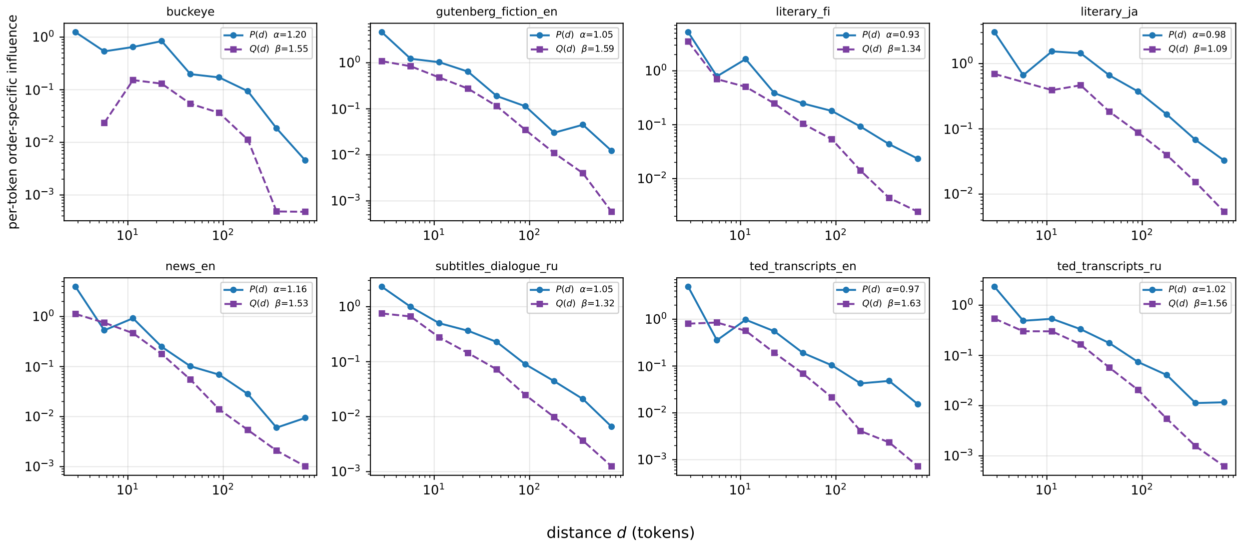

Figure S1: Span-internal order contribution $Q(d)$ and range-matched marginal persistence $P(d)$, per corpus. For each of the eight corpora with both estimates (Llama-3.1-8B), the increment-shuffle band-local order function $Q(d)$ and the marginal persistence function $P(d)$ on log–log axes. $Q(d)$ is positive throughout but steeper: the band-local exponent β exceeds the range-matched integrated exponent α in every corpus. Per-corpus fits and values are in `results/exp2_increment_shuffle/qd_summary.csv`.

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